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An industrial carbon block instance segmentation algorithm based on improved YOLOv8

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Automatic cleaning of carbon blocks based on machine vision is currently an important aspect of industrial intelligent applications. The recognition of carbon block types and center point localization are the core contents of this task, but existing instance segmentation algorithms perform poorly in this task. This paper proposes an industrial carbon block instance segmentation algorithm based on improved YOLOv8 (YOLOv8-HDSA), which achieves highly accurate recognition of carbon block types and edge segmentation. YOLOv8-HDSA designs a Selective Reinforcement Feature Fusion Module (SRFF) that utilizes Hadamard product and dilated convolution to enhance the feature representation of carbon block regions and suppress background noise, fully utilizing the complementary advantages of semantic and detail information to enhance feature fusion capabilities. YOLOv8-HDSA adds a convolutional self-attention mechanism with residual structure to the head, preserving important local information of carbon blocks and improving the ability to extract fine-grained edge details and global features of carbon blocks. YOLOv8-HDSA introduces Focaler-IoU as a loss function to dynamically adjust sample weights to optimize regression performance. The experimental results showed that YOLOv8-HDSA improved the average recognition accuracy of carbon blocks by 7.2% and the segmentation accuracy by 3.8% on real industrial datasets.

Keywords Industrial automation, Machine vision, Carbon block instance segmentation, YOLO, Reinforce feature fusion

Anode carbon block is the main material used in aluminum electrolysis, and its quality directly affects the technical performance indicators of the electrolysis process[1]. The surface of the carbon blocks fired by the calcination furnace is coated with impurities such as clay particles, which seriously affect their quality and require the removal of surface impurities. Robots equipped with computer vision can efficiently and accurately remove these impurities from carbon blocks with industrial automation advances. The first step in the cleaning process is to obtain an image through a monocular camera, and then identify the carbon block in the image, extract its type and center point coordinates. The second step is to combine the calibration algorithm to convert the obtained coordinate information into three-dimensional coordinates in the actual environment[2]. The last step is for the robotic arm to efficiently and automatically clean each carbon block based on its type and three-dimensional coordinates. The use of cameras to identify carbon block types and segment carbon block instances in industrial scenarios is the foundation for achieving automatic cleaning, as can be seen from the above process. How to design high-precision carbon block recognition and segmentation algorithms has become a research focus in today's factory carbon block cleaning work.

Traditional image segmentation methods rely on significant differences in color or texture, such as threshold segmentation[3,4], region growing[5,6], and edge detection[7–9]. The gray appearance of carbon blocks and their texture similarity with the surrounding environment make it difficult for these methods to achieve high-performance accuracy in carbon block recognition and segmentation in the industrial field. The semantic segmentation algorithm[10–13] has high accuracy and computational complexity. The interference noise present in carbon block images reduces the accuracy of semantic labels, such as irregular edges covered with impurities on the surface of carbon blocks and high background noise caused by complex industrial sites. The instance segmentation algorithms separate foreground objects from the background and assign each instance its own mask. This

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type of method has rich geometric shape and boundary information, and is suitable for high-precision object segmentation and recognition in fuzzy boundary environments.

The instance segmentation algorithms can usually be divided into two categories: one-stage algorithms and two-stage algorithms[14]. The one-stage algorithms generate segmentation results directly from images in an end-to-end manner. It achieves accurate segmentation of targets by directly predicting their boundaries and masks, and has high computational efficiency. YOLACT[15] decomposes the segmentation task into two parts: one is to generate the original target mask through the network, the other is to generate the coefficients for each target mask. YOLACT improves the efficiency of object detection and segmentation through multi task learning, and can achieve fast real-time segmentation. BlendMask[16] enhances the accuracy and stability of segmentation mask generation through an improved feature fusion strategy, especially exhibiting good robustness when facing complex backgrounds. The two-stage instance segmentation algorithm generates target candidate regions and performs fine segmentation within these regions. Mask R-CNN[17] adds a branch in Faster R-CNN[18] to predict the binary mask of the target for end-to-end object detection and segmentation. The computational complexity of Mask R-CNN is high, which may limit its processing speed in certain real-time scenarios. Cascade R-CNN[19] improves the final segmentation accuracy by cascading multiple Mask R-CNN models to process segmentation results more finely at each stage of the network. These classic instance segmentation algorithms have high accuracy and high computational cost, making it difficult to meet the needs of enterprises for high-precision, low-cost, and real-time carbon block cleaning equipment.

In recent years, researchers have mainly focused on improving processing speed, segmentation efficiency, and generalization ability in various tasks and environments for instance segmentation algorithms in industrial production[20]. Fang[21] simplified the model structure by introducing a new query mechanism called QueryInst, which transforms instance segmentation into a query task for image regions to solve the real-time segmentation challenge in industrial tasks. Cheng[22] developed a sparse proposal network that improves computational efficiency and speed by minimizing redundant calculations of irrelevant information during detection and segmentation. These algorithms handle targets with rich details and highly similar background features, such as carbon blocks, the segmentation accuracy significantly decreases. Cheng[23] introduced Mask2Former based on Transformer architecture, which uses mask attention mechanism to improve representation efficiency and segmentation performance in image segmentation tasks. Tian[24] designed CondInst, which generates dynamic convolution kernels through conditional convolution to adapt to varying instance characteristics with the growing diversity of industrial production demands. This method eliminates the need for large frameworks to achieve a more flexible segmentation process. Yang[25] developed Group R-CNN, which improves instance segmentation through grouping strategy and is more effective in multi object segmentation tasks.

These explorations still have some shortcomings in improving recognition accuracy and segmentation precision in industrial environments. (1) Many algorithms have significantly improved recognition and segmentation accuracy in simple specific tasks, their performance is not satisfactory in complex environments where the target and background are highly similar. (2) These algorithms have not fully considered factors such as strong light, shadow, reflection, and occlusion in industrial environments. Their recognition and segmentation accuracy and stability have limited performance in practical applications.

The instance segmentation algorithms in the YOLO series have low hardware performance requirements and are suitable for real-time high-precision applications[26], especially the YOLOv8 algorithm. Research is gradually focusing on the application of YOLOv8 instance segmentation algorithm in industrial environments to solve common industrial problems such as complex backgrounds, environmental lighting, and object occlusion. Wang[27] proposed an improved YOLOv8s instance segmentation model aimed at solving the problem of real-time recognition and segmentation of stacked parts. This model adopts PoolFormer as the backbone network, and introduces CARAFE upsampling module to improve the edge segmentation performance and background information filtering ability of the stacking part. Liu[28] proposed a lightweight coal dust particle group instance segmentation method based on YOLOv8 to solve problems in images, such as irregular edges of coal powder particles, object attachment, and difficulty in segmenting small objects. This algorithm enhances feature extraction capability by introducing cross stage local dynamic snake convolution, designs an adaptive weight feature fusion path, and optimizes the computational complexity of the model.

This paper proposes a carbon block instance segmentation algorithm YOLOv8-HDSA suitable for industrial scenarios to solve the problem of low accuracy in carbon block recognition and segmentation in industrial environments. The improved algorithm has high precision, which can assist robot arms in completing automatic carbon block cleaning tasks, avoid production line stagnation and personnel rework caused by recognition errors, improve production line efficiency, reduce production costs, and liberate human resources from dangerous and arduous tasks.

The main contributions of this study are as follows:

1) A Selective Reinforcement Feature Fusion (SRFF) module based on Hadamard[29] is designed to replace the Concat operation in the feature fusion stage to enhance the fusion ability between shallow and deep network features.
2) The ConvSA module is integrated into the Head structure to capture rich global information while preserving the local edge features of carbon blocks. This algorithm reduces the interference of environmental noise on recognition and segmentation, and improves the instance segmentation accuracy of carbon blocks.
3) The Focaler-IoU is introduced as a loss function to focus on regression samples of different difficulty levels. It dynamically adjusts sample weights to optimize regression performance and reduce the impact of simple sample advantages and insufficient optimization of difficult samples.

Materials and methods

Dataset acquisition

The carbon block images used in this experiment were collected from on-site factories using Hikvision DS-2XE3026FWD-I explosion-proof cylinder network cameras with a resolution of 1280 × 720. These cameras are distributed to the cleaning workshop of the factory for monocular recognition. They are fixed on the production line in the factory workshop and capture various types of carbon blocks passing through the production line from a vertical angle, as shown in Fig. 1 (a). We captured video clips between 08:00 and 17:50 and obtained carbon block images using keyframe extraction techniques, as shown in Fig. 1 (b).

We selected a total of 2085 images representing 9 different types of carbon blocks for segmentation. The annotated dataset is divided into training, validation, and testing sets in a ratio of 7:2:1. The training set contains 1459 images, the validation set contains 417 images, and the test set contains 209 images. One carbon block pass through each cleaning process, and this dataset contains a total of 2085 annotated carbon block instances. Figure 2 shows the number of instances for each type of carbon blocks.

These images reflect the real situation of carbon blocks in industrial production. They contain different characteristics of carbon block and the environmental factors:

(1) The surface of carbon blocks exhibits diverse characteristics due to different treatments during the production process. Different types of carbon blocks have different surface structures and textures, which makes the identification task of carbon blocks highly challenging, as shown in Fig. 3. The same type of carbon block can exhibit different appearance characteristics due to changes in the degree of carbon particle coverage. Figure 4a–c show the low carbon particle coverage, normal carbon particle coverage, and high carbon particle coverage of the same type of carbon block, respectively. Their external features have significant differences, which increases the difficulty of the algorithm in object detection and instance segmentation.

(2) The surface of carbon blocks is highly similar to the texture features of the surrounding environment due to the gray appearance formed during the production process. The polished carbon particles will accumulate on the ground during the automatic cleaning task of carbon blocks. The ground features will gradually resemble the color and texture of the carbon block surface, which increases the background interference in the instance segmentation task, as shown in Fig. 5. The equipment characteristics in the factory workshop may be similar to the carbon block characteristics in areas without carbon particle coverage, as shown in Fig. 6.

These influencing factors often result in lower accuracy of universal instance segmentation algorithms in carbon block recognition and edge segmentation. It is necessary to improve a high-precision instance segmentation algorithm suitable for automatic carbon block cleaning tasks in industrial environments based on existing algorithm research.

Methods

Structure diagram of YOLOv8n-seg and YOLOv8-HDSA

The network structure of the YOLOv8n-seg model can be divided into three main parts: Backbone, Neck, and Head. The Backbone network CSPDarknet improves the model's feature extraction capability, and can accurately extract carbon block features and background features in carbon block instance segmentation tasks. we continue to serve it as a Backbone network structure for improving algorithms. YOLOv8n-seg's Neck uses Concat operation for feature fusion. This can easily lead to feature confusion in carbon block instance segmentation tasks, and improvement is needed to enhance multi-scale feature fusion and context awareness capabilities. Carbon blocks have issues with fuzzy boundaries and significant intra class differences in dataset images. The original Head structure has low segmentation accuracy when dealing with such problems, and its ability in object classification, boundary prediction, and detailed segmentation needs to be improved. We are mainly committed to optimizing and improving Neck and Head components, as shown in Fig. 7a, b.

This study optimized the neck and head of YOLOv8n-seg and proposed an improved algorithm YOLOv8-HDSA suitable for automatic carbon block cleaning tasks, as shown in Fig. 7(b):

(a) Carbon block cleaning site **(b) Nine types of carbon block images**

Fig. 1. Dataset acquisition.

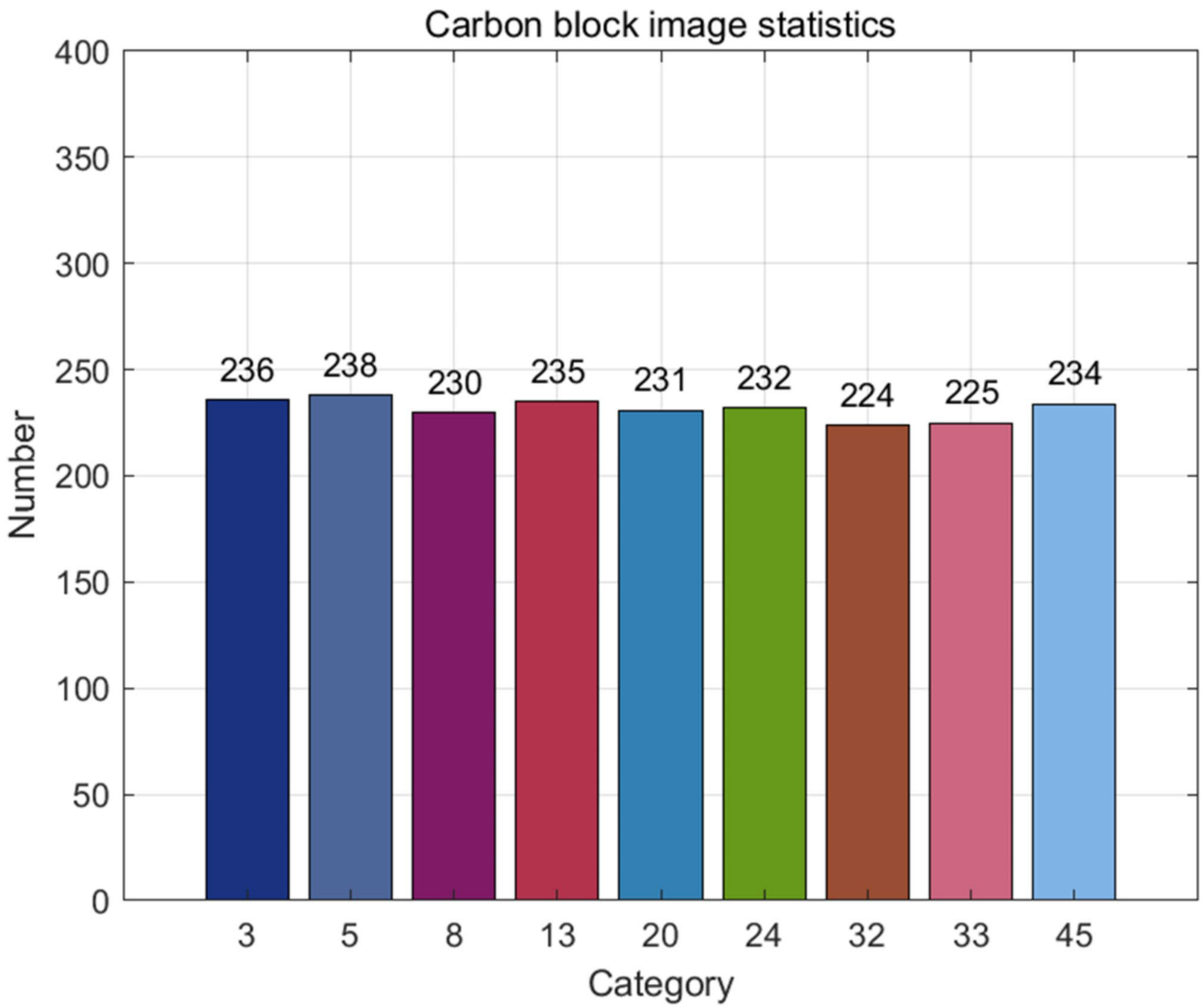

Fig. 2. The number of instances for each type of carbon blocks.

(1) The backbone network of the model is consistent with YOLOv8n-seg. The input resolution is 640 × 640 resolution image, and three types of feature maps with resolutions of 80 × 80, 40 × 40, and 20 × 20 are output to enter the feature fusion network structure.

(2) Neck uses Concat operation to achieve multi-scale feature fusion in the traditional YOLOv8n-seg structure. We have designed a Selective Reinforcement Feature Fusion (SRFF) module to replace all Concat, Upsampling, and Downsampling operations. In the bottom-up part, the input of the first SRFF module is the feature map output by the feature pyramid module (SPPF) with a resolution of 20 × 20 and the feature map output by the sixth layer of the backbone network with a resolution of 40 × 40. The output is a 40 × 40 feature map with 512 channels. This feature map is input into the second SRFF module after the C2f module and fused with the 80 × 80 feature map output from the fourth layer of the backbone network. The output is an 80 × 80 feature map with 256 channels. In the top-down section, the first SRFF module inputs the 80 × 80 feature map output from the previous C2f module and the 40 × 40 feature map output from the first C2f module in the bottom-up section. After fusion, a 40 × 40 feature map with 512 channels was obtained. It is input into the last SRFF module after the C2f module and fused with the 20 × 20 feature map output by the feature pyramid module to obtain 1024 channels of 20 × 20 feature maps.

(3) This module can capture a wider range of contextual information while preserving local information and reducing computational complexity. It can improve the predictive ability of the three neck feature maps without changing the size of the feature map space and the number of channels.

Selective reinforcement feature fusion (SRFF) module

The Neck structure of YOLOv8n-seg is shown in Fig. 8. The Concat operation is used as the feature fusion module. The fixed Upsampling and Downsampling layers are used to ensure that the spatial resolution of feature maps at different scales is the same. The essence of Concat operation is to connect the features of each layer in the channel dimension without filtering or adjusting the weights of the features. This leads to various interference

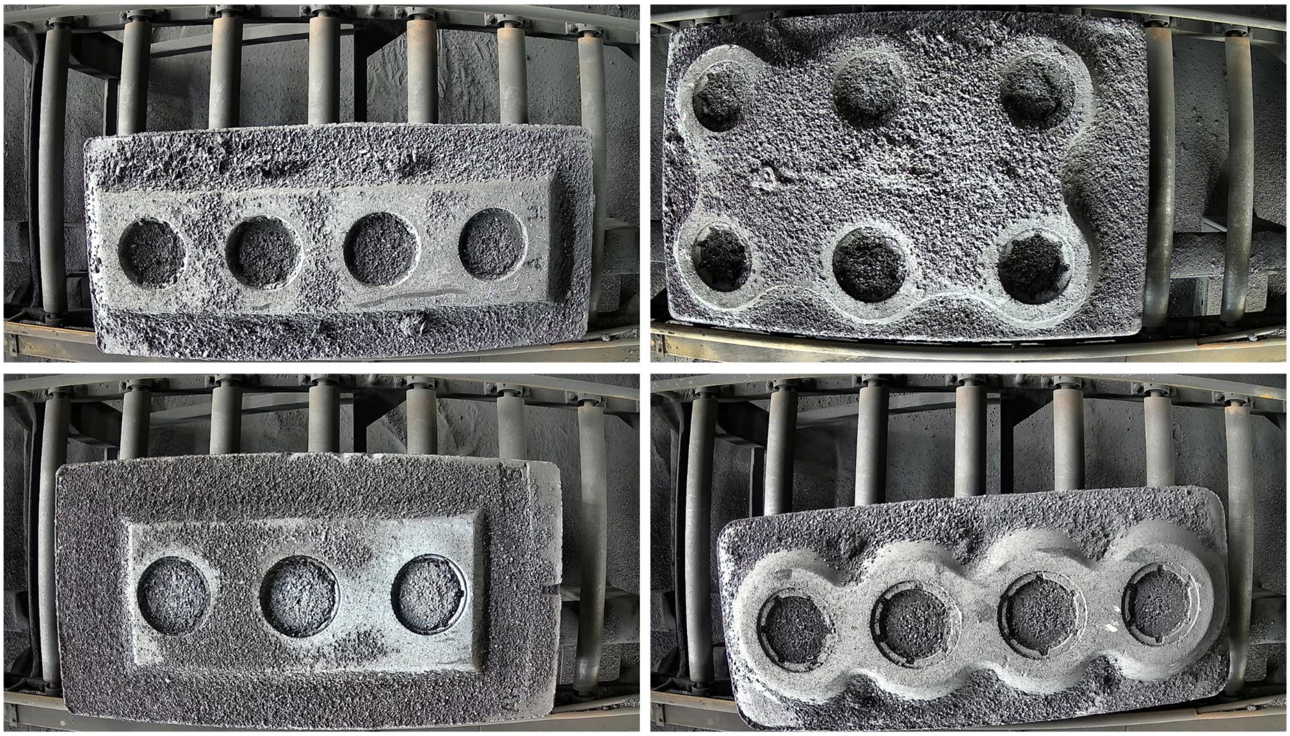

Fig. 3. Characteristic differences between different types of carbon blocks.

(a) Low carbon particle **(b) Normal carbon particle** **(c) High carbon particle**

Fig. 4. Characteristic differences between carbon blocks of the same type. (**a**) Low carbon particle (**b**) Normal carbon particle (**c**) High carbon particle.

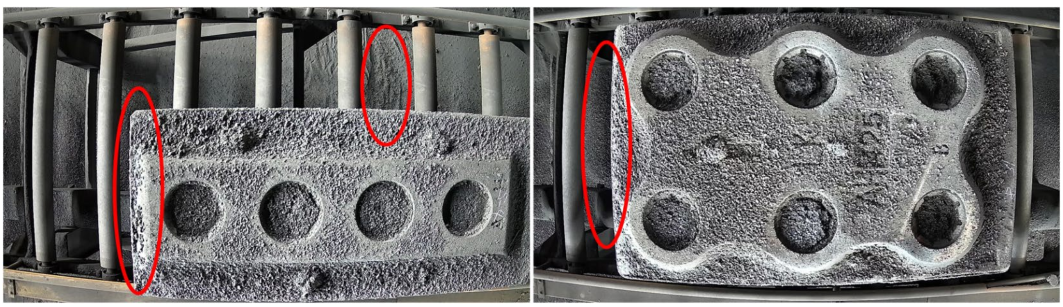

Fig. 5. Characteristic similarity between carbon blocks and ground.

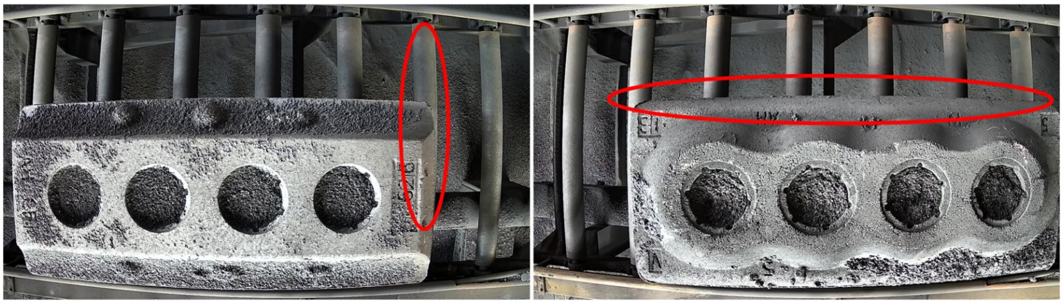

Fig. 6. Characteristics similarity between carbon blocks and mechanical structures.

signals in the fused feature map, resulting in feature redundancy and information mixing. Concat operation can easily introduce environmental noise, which can affect the accuracy of edge carbon block segmentation. The fixed scaling method cannot optimize the specific features of the target in the input image when carbon blocks have large-scale variations and strong background interference. The fixed Upsampling and Downsampling layers may introduce interpolation errors, resulting in the loss of some detailed information in the input feature map (such as blurry carbon block boundaries or background), which affects subsequent feature fusion.

We propose the Selective Reinforcement Fusion Feature (SRFF) module to address this issue and improve the Neck structure, as shown in Fig. 9. The feature map output by SPPF is processed through four SRFF and four C2f modules to generate the final predicted mask. This improvement eliminates separate Upsampling and Downsampling layers, which allow the SRFF module to flexibly adjust feature map sizes using adaptive average pooling and bilinear interpolation to preserve important details. This enhances the network's ability to handle multi-level features and improves adaptability to different input conditions.

Figure 10 shows that the SRFF structure consists of three parts: spatial alignment layer, Hadamard layer, and extended convolutional layer. The spatial alignment layer includes adaptive average pooling and bilinear interpolation.

When the spatial size of the deep feature map is smaller than that of the shallow feature map. SRFF adjusts the size of the depth feature map based on the selection of bilinear interpolation, as shown in formula (1):

$$Y_{i,j} = \frac{1}{N} \sum {}_{m=0}^{H_{in}-1} \sum {}_{n=0}^{W_{in}-1} X_{m,n} \tag{1}$$

where $Y_{i,j}$ is the pixel value of the output feature map. $X_{m,n}$ is the pixel value of the input feature map. N is the number of pixels within the specified output area. H_{in} and W_{in} is the height and width of the input feature map. This formula will automatically calculate the suitable area for the target output size to ensure that the size of the output feature map is H_{out} and W_{out} .

When the spatial size of deep feature maps is larger than that of shallow feature maps. SRFF uses adaptive average pooling to adjust the size of the depth feature map, as shown in formula (2):

$$F_{out}\left(i,j\right) = \frac{1}{M_h \times M_w} \sum {}_{m=0}^{M_h-1} \sum {}_{n=0}^{M_w-1} F_{in}\left(h_i+m, w_j+n\right) \tag{2}$$

where $F_{out}\left(i,j\right)$ is the value at position $\left(i,j\right)$ of the output feature map. $F_{in}(h_i+m, w_j+n)$ is the value at position $\left(h_i+m, w_j+n\right)$ of the input feature map. M_h and M_w are the height and width of the input feature map that contribute to the average value at each position of the output feature map. h_i and w_j are the indices derived from evenly dividing the input feature map.

The Hadamard layer mainly includes two operations: convolutional feature extraction of feature maps and element wise multiplication. This layer uses convolutional kernels with a stride of 1 and a size of 3×3 to extract local features and perform element wise multiplication operations. This operation aims to enhance the important features of carbon blocks extracted from convolution while suppressing irrelevant background features. As shown in reference[30], formula (3):

$$C_{ij} = A_{ij} \cdot B_{ij} \tag{3}$$

where C is the resulting matrix, A_{ij} and B_{ij} are the elements of matrices A and B at the i-th row and j-th column, respectively. Any two $m \times n$ matrices A and B, the result C will also be an $m \times n$ matrix.

The SRFF structure introduces a dilated convolutional layer to compensate for the loss of detail information that may occur during the process of resizing feature maps. This layer uses a 3×3 dilated convolution kernel with an expansion ratio of 4 and a filling ratio of 4 to expand the receptive domain to 7×7 while maintaining the spatial resolution of the feature map. It further integrates the fused carbon block features to improve the robustness and accuracy of the model in carbon block cleaning tasks.

Convolutional self-attention mechanism with residual structure (ConvSA) module

The Head obtains feature maps of three different spatial sizes and channel numbers from the neck of the model, 80×80×256, 40×40×512, and 20×20×1024. The Head of the Yolov8n-seg model (Fig. 11a) consists of two

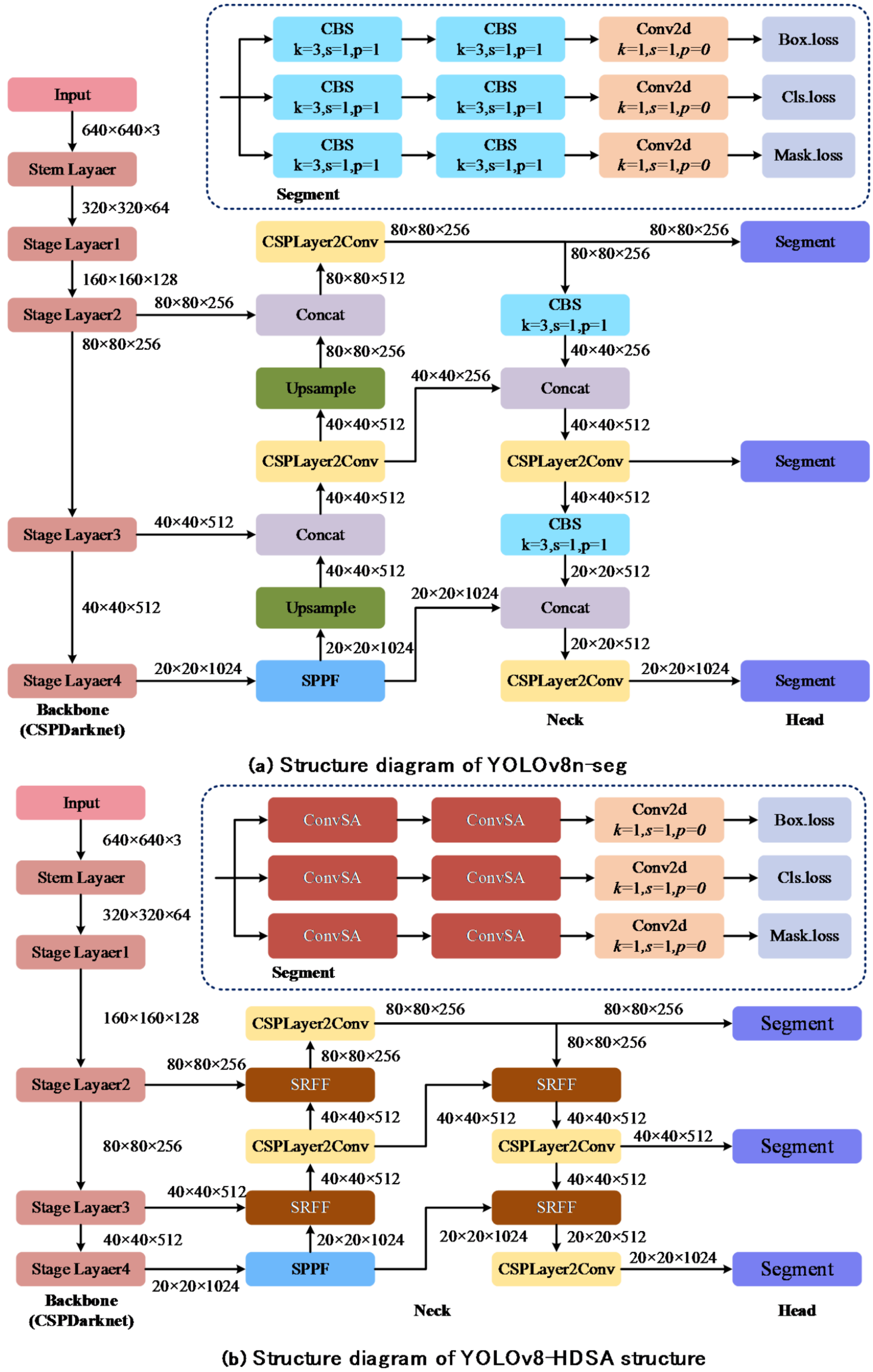

Fig. 7. Structure diagram of YOLOv8n-seg and YOLOv8-HDSA.

CBS convolution modules (Fig. 11b) and a 1 × 1 convolution operation. This Head structure lacks an explicit mechanism for integrating multi-scale information and relies solely on stacking feature maps followed by further convolutions, which is less effective in efficiently capturing multi-scale features of complex targets. Convolution module mainly extracts features through local receptive fields. It has a certain degree of non-linear expression ability, and is difficult to capture the contextual dependencies at long distances in the input feature map, which can easily lead to detail loss and background noise introduction in carbon block cleaning tasks.

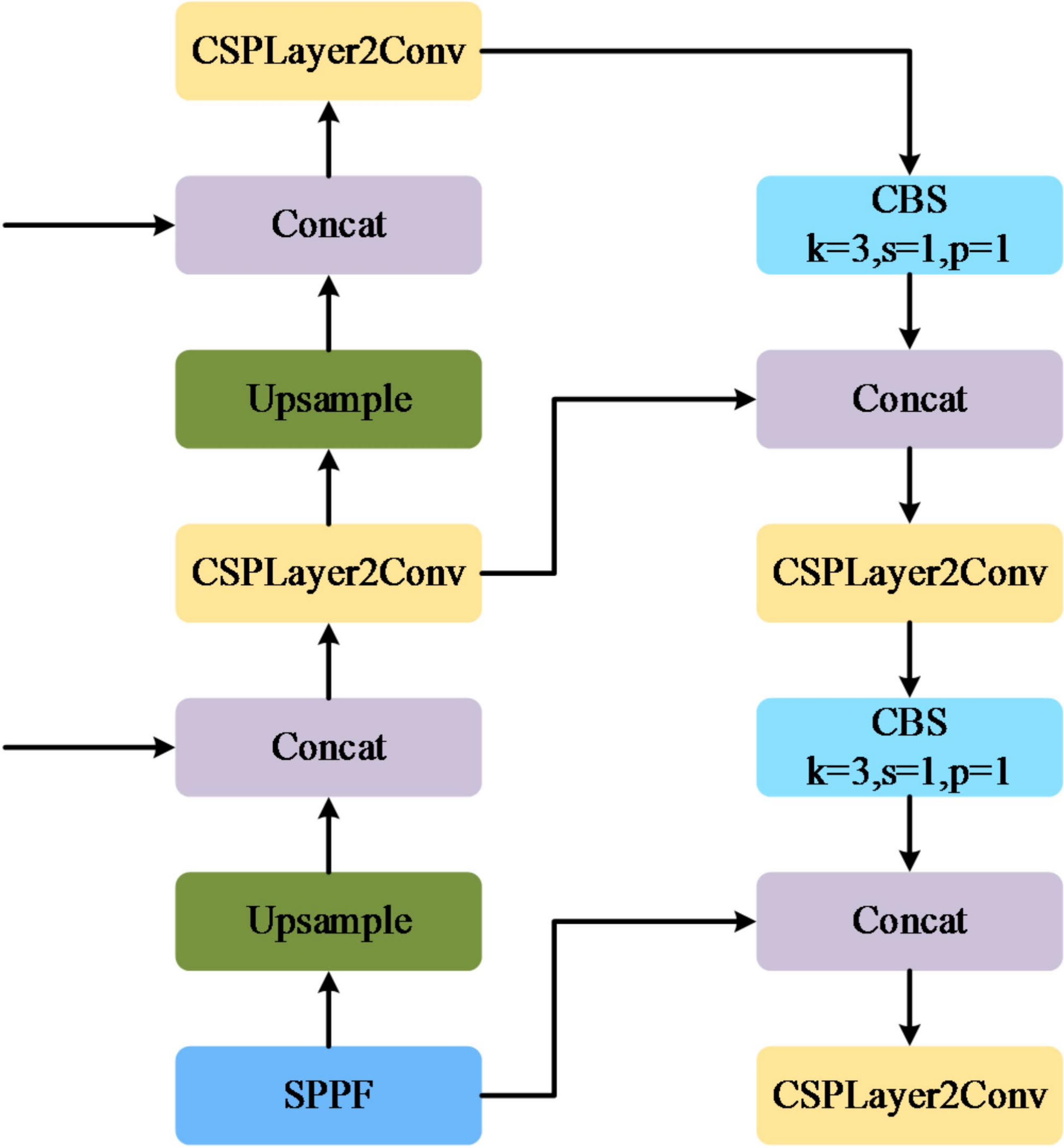

Fig. 8. The Neck structure of YOLOv8n-seg.

We have designed a convolutional self-attention mechanism with a residual structure (ConvSA) module, which replaces the CBS module without changing the input and output of the original structural feature map, which enhance the adaptability of the model to the actual application environment of carbon blocks and improve the accuracy of recognition and segmentation. Figure 12 (a) shows that the output path structure of feature maps with different spatial sizes is consistent. These structures are generated through ConvSA module and 1×1 convolution operation to generate three types of feature maps for predicting Box, Cls, and Mask.

We designed the ConvSA module as shown in Fig. 12 (b). It effectively integrates convolution operations, self-attention mechanisms, and residual connections. This module includes a backbone path[31] and a parallel convolution module to allow the model to extract global features while preserving the original feature information. The backbone path starts from the CBS module, which uses a 1×1 convolution kernel to compress the number of channels and performs batch normalization (BN) and activation function (SiLU) to enhance feature nonlinearity. The output feature map is input into the self-attention module to calculate the relationships

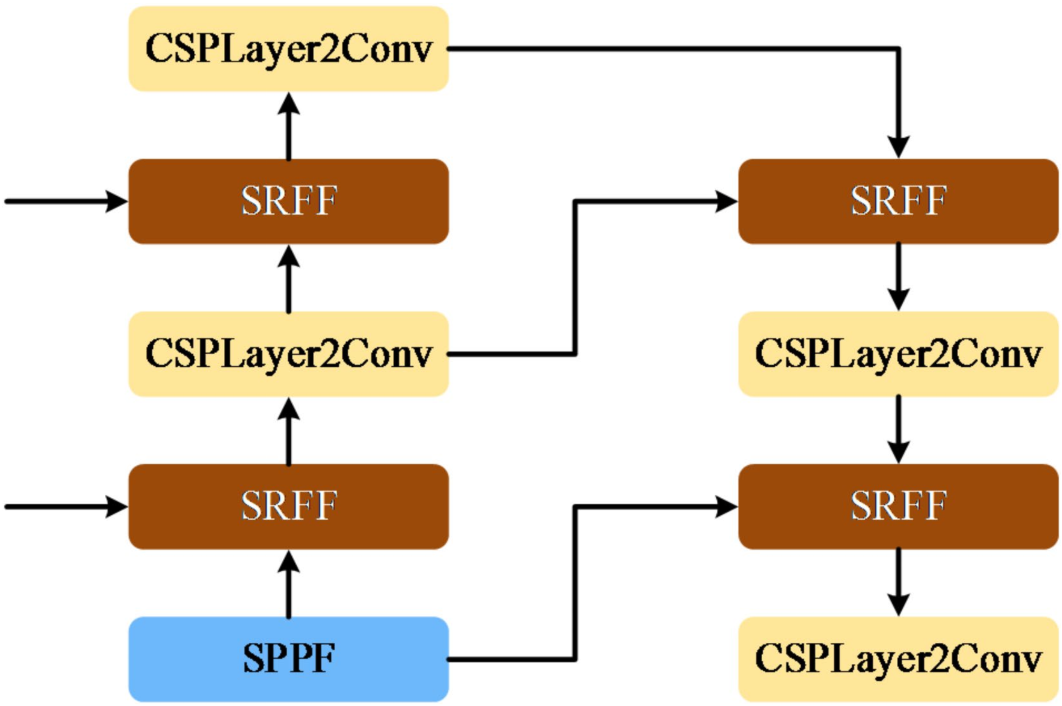

Fig. 9. The Neck structure of YOLOv8-HDSA.

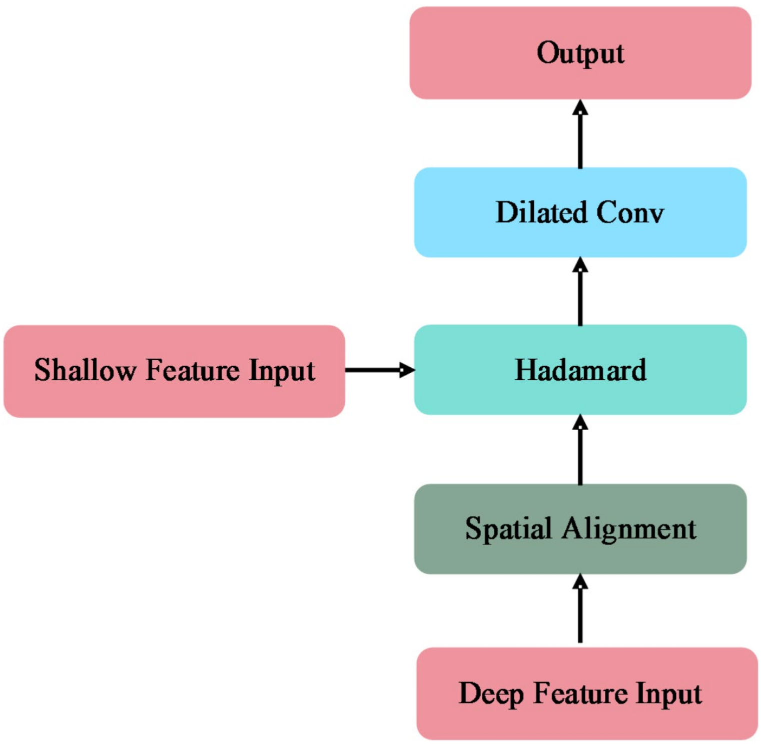

Fig. 10. Application of SRFF module.

between all pixel pairs in the input feature map, enabling the model to focus on the global relationships between different regions in the carbon block image. The 1×1 convolution module processes the feature maps output by the self-attention mechanism, converting the number of feature channels generated by the self-attention mechanism into dimensions that match the input features. Further enhance the extraction of global features of carbon blocks.

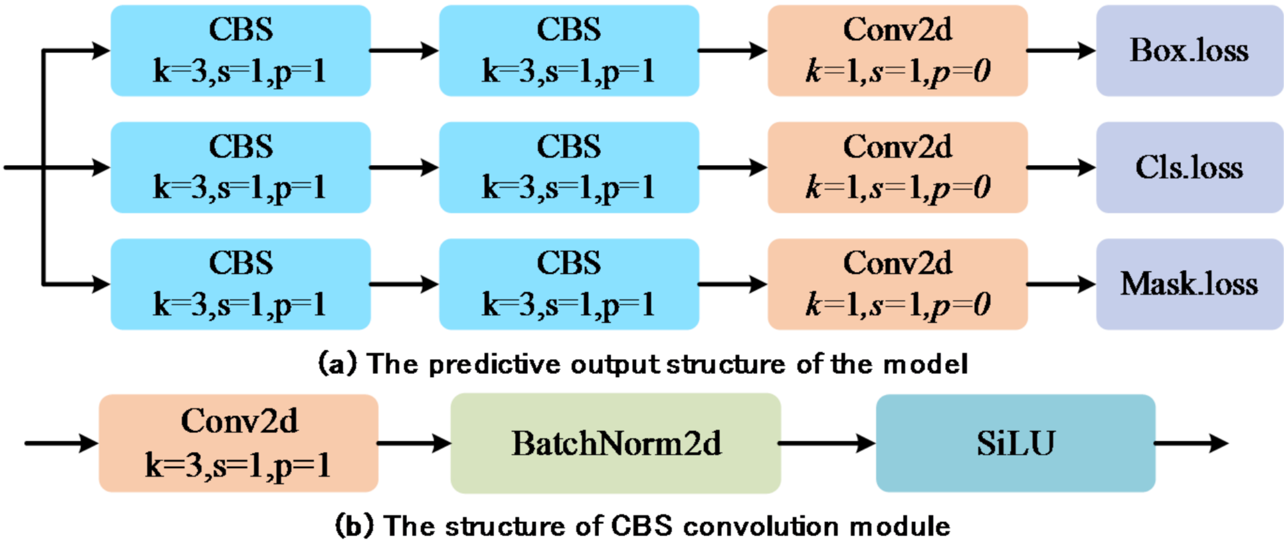

Fig. 11. The Head structure of YOLOv8n-seg.

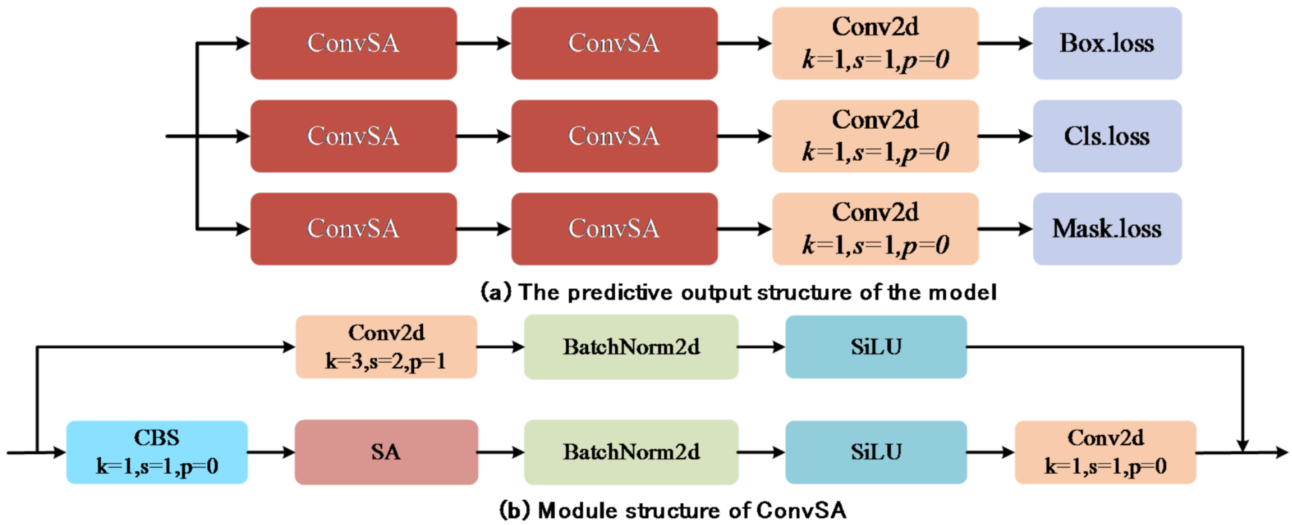

Fig. 12. The Head structure of YOLOv8-HDSA.

The introduction of self-attention mechanism has the following limitations: (1) The edges and details of carbon blocks may be overlooked when focusing too much on global information, and these features are crucial for distinguishing carbon block types and segmenting carbon block edges. (2) The global perspective of self-attention may make the shape, size, and height of the carbon block surface smoother on the feature map, which reduces the importance of local information and makes it difficult to distinguish subtle differences between similar carbon blocks. We introduce residual structures to preserve local feature representations as a solution to this problem. Figure 12 (b) shows the improved structure, where the original input feature map is added to the main path output through parallel paths. Convolution operation (3×3, stride 2, padding 1) extracts local features of carbon blocks. After batch normalization and activation function, the output feature map is added to the feature map of the main path. This skip connection preserves the details and semantic information in the original feature map, which avoids the decrease in local feature expression ability caused by the self-attention mechanism.

Focaler-IoU loss function

There are many actual samples, some of which are difficult for the model to accurately predict in the carbon block instance segmentation task due to the high similarity between the target and background. These types of samples are usually considered difficult samples because there is a large error between their predicted values and the true values. If such samples are not processed, it may lead to overfitting of the model when facing complex samples. Simple samples are samples that the model can easily and accurately predict, which means that the error between the predicted value and the true value is very small. They exhibit significant boundary differences between carbon blocks and the background, as well as notable features of the carbon blocks themselves, in the task of segmenting carbon block instances. The model's excessive focus on simple samples can lead to insufficient learning of difficult samples, which can reduce overall performance.

The YOLOv8n-seg model uses CIoU (Complete Intersection over Union) as the loss function. This loss function effectively focuses on the geometric relationship between bounding boxes, with less attention paid to the distribution of difficult and simple samples, which cannot handle the weight changes between difficult and simple samples. This study introduces the Focaler-IoU loss function to dynamically adjust the focus on different samples. This will reduce the impact of difficult and simple sample distributions on model training.

The Focaler-IoU reconstructs IoU loss through linear interval mapping to achieve attention to different samples. The complex background, highly similar features, and occlusion in the carbon block dataset result in a higher proportion of difficult samples. The model's inability to effectively optimize these samples can lead to inaccurate segmentation boundaries or object detection failures. The Focaler-IoU optimizes these samples in a targeted manner through a dynamic weighting mechanism to improve the robustness of the model. The contribution of simple samples to the model gradient tends to zero in the later stages of training, and continuing to optimize these samples will result in a waste of model resources. The Focaler-IoU reduces the focus on simple samples to free up more computing resources at this stage. As shown in reference[32], formula (4):

$$IoU^{\text{focaler}} = \begin{cases} 0\,, & IoU < d \\ \frac{IoU-d}{u-d}\,, & d \ll IoU \ll u \\ 1\,, & IoU > u \end{cases} \tag{4}$$

This formula adjusts the loss based on the value of the intersection to union ratio (IoU). When IoU is less than a lower threshold d, the loss is 0; when the IoU is greater than an upper threshold u, the loss is 1; when the IoU is between d and u, the loss is a function that increases linearly based on the IoU value.

This design allows the loss function to be sensitive to IoU values within a certain range. This allows the model to focus more on samples with moderate overlap between the predicted bounding box and the real bounding box, i.e. samples that are neither too difficult nor too easy. This helps the model better learn to extract features from moderately difficult samples, rather than just focusing on the simplest or most difficult samples.

Results

Experimental setup

The model improvements and experiments were conducted on a server with the configuration shown in Table 1. The experiment used GPU hardware and CUDA 11.6 to accelerate calculations and improve computational efficiency. The parameter settings for models with different structures are unified. Each training batch contains 16 images, with input image resolution set to 640×640 pixels, the number of epoch is 400. Hyperparameter settings used: initial learning rate is 0.01, final learning rate is 0.01, learning momentum is 0.937, weight_ Decay is 0.0005.

Evaluation criteria

Common performance metrics for evaluating instance segmentation algorithms include accuracy, recall, mAP (mean accuracy), parameters (model parameter count), and FLOP (floating point operations per second).

In this paper, the evaluation task of the model is divided into the accuracy of identifying carbon block types and the accuracy of pixel segmentation of carbon block areas, in response to the demand for carbon block cleaning tasks in industrial scenarios. The mAP of Box and Mask were selected as the evaluation criteria for the model. The mAP represents the mean precision across all classes in the segmentation results, and its calculation is shown in formula (5):

$$\begin{aligned} AP &= \sum_{i=1}^{n-1} (r_{i+1} - r_i)\, p(r_{i+1}) \\ mAP &= \frac{1}{m} \sum_{i=1}^{m} AP_i \end{aligned} \tag{5}$$

In the formula (5), n represents the total number of positive samples during model validation, r_i refers to the Recall value in ascending order, and $p\,(r_i)$ represents the Precision value corresponding to a Recall value of r_i, and m represents the number of classes present during the model validation process, P_i refers to the Precision

Experimental machine environment	Experimental compilation environment
CPU: Intel(R)Xeon(R)Bronze 3206R CPU @ 1.90GHZ	Python 3.9.19
GPU: Nvidia Quadro RTX 4000 16G	CUDA 11.6
OS: Windows 11	Pytorch 1.12.0

Table 1. Experimental machine configuration and compilation environment.

values in ascending order, and r_i indicates the Recall value corresponding to each Precision value. The values in mAP represent IOU thresholds.

The mAP@0.5 calculate the AP for all categories and take the average when the IOU threshold is 0.5. The calculation is shown in formula (6):

$$mAP@0.5 = \frac{1}{m}\sum_{i=1}^{m} AP_i^{IOU=0.5} \tag{6}$$

In the formula (6), IOU = 0.5 means using only an IOU threshold of 0.5 to calculate AP for each category.

The mAP@0.5 0.95 is used to calculate the AP of all categories at multiple IOU thresholds (ranging from 0.5 to 0.95, with a step size of 0.05) and take the average. The calculation is shown in formula (7):

$$mAP@0.5:0.95 = \frac{1}{T}\sum_{t=1}^{T}\frac{1}{m}\sum_{i=1}^{m} AP_i^{IOU=t} \tag{7}$$

In the formula, T is the number of IOU thresholds. $AP_i^{IOU=t}$ represents the AP of class i when the IOU threshold is t.

As the IOU threshold increases, the overlap requirement between predicted and ground truth boxes becomes stricter, making it more difficult to achieve higher values.

Accurate carbon block edge segmentation is crucial for subsequent automatic cleaning. We must evaluate the segmentation accuracy, generalization ability, and edge detail processing effect of the model from different perspectives. Therefore, we choose mAP@0.5 and mAP@0.5:0.95 to evaluate the instance segmentation accuracy of the algorithm.

Results and analysis

Experiments were conducted to compare YOLOv8-HDSA, YOLOv8n-seg, and other mainstream instance segmentation algorithms, including Mask R-CNN, YOLACT, SOLOv2, BlendMask, CenterMask, YOLOv5n-seg, and YOLOv7n-seg. The results are shown in Table 2.

It can be seen that the classic instance segmentation algorithm Mask R-CNN did not exceed 80% in various metrics. The performance of SOLOv2, YOLACT, CenterMask, and BlendMask has improved, but the accuracy of Box and Mask has not exceeded 90%.

We can see that the YOLO series of methods such as the YOLOv5n-seg, YOLOv7n-seg, and YOLOv8n-seg exhibit high accuracy. YOLOv7n-seg is in Box(mAP@0.5:0.95) and Mask(mAP@0.5: 0.95) achieved 92.1% and 95.2% respectively, which is the optimal method for automatic carbon block cleaning tasks in industrial scenarios previously applied. This paper proposed YOLOv8-HDSA model achieved breakthrough performance improvements in all metrics, with an average recognition and segmentation accuracy of 99.5%. The recognition and segmentation accuracy under high IOU reached 99.3% and 99.4%, respectively, which were 7.4% and 7.7% higher than YOLOv8n-seg, 7.2% and 3.8% higher than YOLOv7n-seg.

Image analysis of carbon blocks of the same type

Figure 13 shows the detection results of YOLOv7n-seg, YOLOv8n-seg, and YOLOv8-HDSA among carbon blocks of the same type in the test set. We have selected three images of type 5 to represent carbon blocks of the same type with different degrees of carbon particle coverage. Different levels of carbon particle coverage may result in different target features between carbon blocks of the same type, which can affect the recognition and segmentation accuracy of the model. As shown in Fig. 13 (d), Fig. 13 (e), Fig. 13 (f), Fig. 13 (g), Fig. 13 (h), and Fig. 13 (i), YOLOv7n-seg and YOLOv8n-seg correctly identified the carbon block type and maintained high confidence, their output mask maps showed a large number of mask overflow and omission areas at the edge of the carbon block. As shown in Fig. 13 (j), Fig. 13 (k), and Fig. 13 (l), the improved model exhibits higher confidence than the previous two models in accurately identifying carbon block types. There are a large number of mask overflow and missing areas in YOLOv7n-seg and YOLOv8n-seg, and the improved model achieves precise mask coverage. These improvements indicate that our method can maintain high recognition accuracy and precise segmentation even if there are strong feature differences between the same type.

Method	Box (mAP@0.5)	mAP@0.5:0.95	Mask (mAP@0.5)	mAP@0.5:0.95
Mask R-CNN	79.5	78.3	79.5	77.6
SOLOv2	83.4	82.4	83.4	83.7
BlendMask	85.6	84.1	85.6	85.2
CenterMask	88.1	83.3	88.1	85.7
YOLACT	88.9	87.2	88.9	87.5
Yolov5n-seg	90.5	88.1	90.5	90.4
Yolov7n-seg	96.4	92.1	96.4	95.2
Yolov8n-seg	92.2	91.9	92.2	91.1
Yolov8-HDSA	99.5	99.3	99.5	99.4

Table 2. Comparison of model segmentation accuracy.

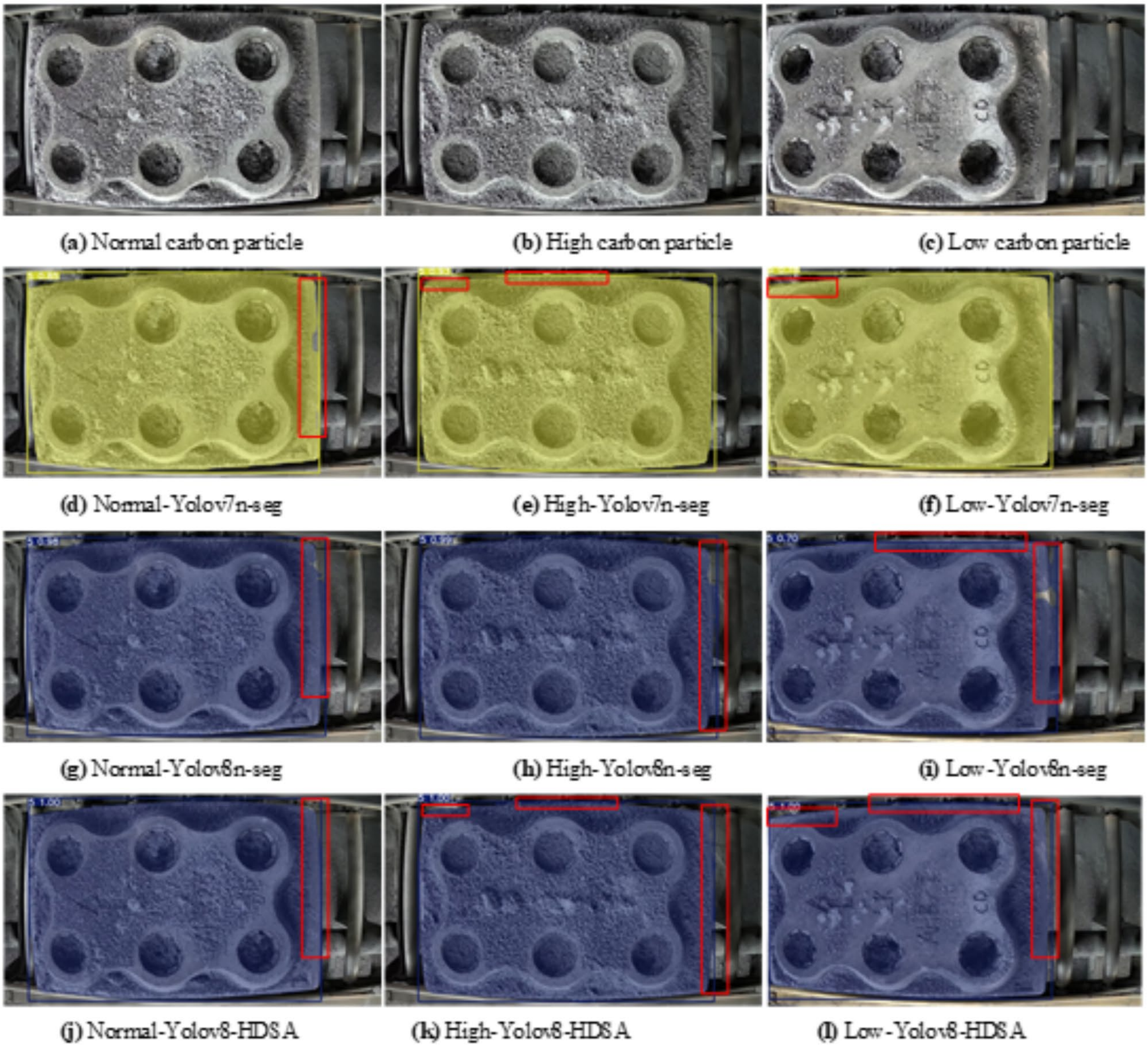

Fig. 13. Comparison of YOLOv7n-seg, YOLOv8n-seg and YOLOv8-HDSA effects for same type of carbon blocks.

Image analysis of carbon blocks of different types

Figure 14 shows the predicted output results of YOLOv7n-seg, YOLOv8n-seg, and YOLOv8-HDSA among different types of carbon blocks. We have selected images with normal carbon particle coverage for each type of carbon block, which have the highest number of carbon block instances in actual production. From Fig. 14, it can be seen that YOLOv7n-seg and YOLOv8n-seg have misidentification, obvious mask overflow and missing, while the improved model output results do not have the above problems. According to the comparison results in Fig. 14, the improved model can accurately identify and segment all types of carbon blocks in industrial scenarios, capturing subtle changes in details.

Ablation study

The improvement of YOLOv8 HDSA includes three aspects: selective reinforcement feature fusion (SRFF) module, convolutional self-attention mechanism with residual structure (ConvSA) module, and introduction of Focaler-IoU loss function. The ablation experimental results of these methods are shown in Table 3.

The ablation study verified whether the improvement could enhance the performance of the model. From Table 3, it can be concluded that when the SRFF module is introduced separately to replace the Upsampling and Downsampling operations in the neck network, as well as the Concat module, the recognition and segmentation tasks mAP@0.5 All have increased by 3%, mAP@0.5:0.95 increased by 0.9% and 3.3% respectively. From Table 3, the average accuracy of recognition and segmentation tasks is achieved when introducing the ConvSA module separately at the head mAP@0.5 increased by 1.2%, mAP@0.5:0.95 increased by 0.2% and 0.3% respectively. The

Fig. 14. Comparison of YOLOv7n-seg, YOLOv8n-seg, and YOLOv8 HDSA predicted outputs for different types of carbon blocks (first column: original image, second column: YOLOv7n-seg output, third column: YOLOv8n-seg output, fourth column: YOLOv8-HDSA).

SRFF module can effectively improve the fusion ability of deep and shallow network features, while the ConvSA module extracts richer global information. SRFF module and ConvSA module are introduced simultaneously, the recognition and segmentation tasks mAP@0.5 increased by 7.3%, mAP@0.5:0.95 increased by 6.5% and 7% respectively. The Focaler-IoU loss function is used in the prediction output stage to further improve the localization accuracy of carbon block boundaries by reducing the impact of different regression samples on

Method	Box (mAP@0.5)	mAP@0.5:0.95)	Mask (mAP@0.5	mAP@0.5:0.95)
Yolov8n-seg	92.2	91.9	92.2	91.1
Yolov8n-seg + SRFF	95.2	92.8	95.2	94.4
Yolov8n-seg + ConvSA	93.4	92.1	93.4	91.4
Yolov8n-seg+ SRFF + ConvSA	99.5	98.4	99.5	98.1
Yolov8n-seg+ SRFF + ConvSA + Focaler-IoU	99.5	99.3	99.5	99.4

Table 3. Comparison of segmentation accuracy in ablation experiments.

model training. Compared to the previous YOLOv8-HDSA model, the Focaler-IoU loss function improves the model's performance mAP@0.5:0.95 has increased by 1.9% and 1.3% respectively. Based on the data provided in Table 3, it can be concluded that introducing SRFF module and ConvSA module to optimize Neck and Head in YOLOv8n-seg network structure, while using Focaler-IoU loss function to quantify the difference between model prediction results and real labels, can significantly improve the recognition accuracy and segmentation accuracy of carbon blocks.

Conclusions

In this paper, we study the instance segmentation problem of carbon blocks in industrial automatic carbon block cleaning, and find that the high feature similarity between the target carbon block and the background in the image can lead to high misidentification rate and low localization accuracy of mainstream algorithms. We propose the YOLOv8-HDSA carbon block instance segmentation algorithm to solve this problem. YOLOv8-HDSA introduces a Selective Reinforcement Feature Fusion (SRFF) module that utilizes Hadamard product and dilated convolution to facilitate effective fusion of semantic and detail information. This method selectively enhances the features of the target area while suppressing irrelevant background features. We combine residual structure with convolutional self-attention mechanism to design the ConvSA module. This module enhances the capture of global contextual information without losing the ability to express local features. This significantly improves the accuracy of carbon block type recognition and region segmentation. We introduce Focaler-IoU as the loss function to focus on regression samples of different difficulty levels. It dynamically adjusts sample weights to optimize regression performance and reduce the impact of simple sample advantages and insufficient optimization of difficult samples.

We designed multiple comparison and ablation experiments in the dataset to verify the recognition accuracy and segmentation accuracy of the YOLOv8-HDSA algorithm. The experimental results show that the recognition accuracy of this algorithm has improved by 7.2% compared to mainstream instance segmentation algorithms, and the segmentation accuracy has increased by 3.8%.

The YOLOv8-HDSA algorithm cannot recognize the posture changes of carbon blocks. We need further research to expand its ability to recognize and segment different poses of carbon blocks. Future work will continue to improve based on this foundation.

Data availability

The datasets used and analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

R.S. wrote the main manuscript text. Z. L. conducedt manuscript review and revision, and provided leadership and financial support. Z.W. and W.Z conducted experimental design and analysis. Y.X. and G.L. prepared Figs. 1, 2, 3, 4, 5 and 6. P.M. organized experimental data. Z.Z. provided supervision and guidance. All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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